

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science & Engineering

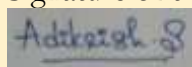
ASSESSMENT TOOL 1 - SCENARIO BASED QUESTIONS

Course Code	DSA0404	Course Title	Fundamentals of Data Science
CO Assessed	CO4 - Scenario Based Questions	Assessment	Assessment Tool 1: Scenario Based Questions
Weightage	40% of CIA	Total Marks	100
CO4 Statement	To apply the statistical inferences such as testing of hypothesis and point estimates for the given data		
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Section/Batch	2024	Date	17 August 2026

Code of Conduct:

I, ADITYA KRISHNAN S/192424069, certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Marks
Understanding of Scenario	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyse the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Question 1:

An online learning platform measures the completion time of a course for 100 students. The mean completion time is 14 days and the standard deviation is 3 days. Construct a 95% confidence interval for the true mean completion time.

Solution

Given:

$$\bar{x} = 14, s = 3, n = 100$$

For a 95% confidence interval,

$$z = 1.96$$

Standard Error,

$$SE = \frac{s}{\sqrt{n}} = \frac{3}{\sqrt{100}} = 0.3$$

Margin of Error,

$$ME = 1.96(0.3) = 0.588$$

$$\therefore, CI = \bar{x} \pm ME = 14 \pm 0.588 = \boxed{(13.412, 14.588) \text{ days}}$$

Answer

The 95% confidence interval for the true mean completion time is 13.412 to 14.588 days.

Question 2:

A hospital surveys 400 randomly selected patients, and 340 report that they are satisfied with the service. Construct a 95% confidence interval for the proportion of satisfied patients.

Solution:

Given:

$$n = 400, x = 340$$

Sample Proportion:

$$\hat{p} = \frac{340}{400} = 0.85$$

For a 95% confidence interval,

$$z = 1.96$$

Standard Error,

$$SE = \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}} = \sqrt{\frac{0.85(0.15)}{400}} \approx 0.01785$$

Margin of Error,

$$ME = 1.96(0.01785) = 0.0350$$

$$\therefore, CI = \bar{x} \pm ME = 0.85 \pm 0.0350 = \boxed{(0.815, 0.885)}$$

or $\boxed{81.5\% \text{ to } 88.5\%}$

Answer:

The 95% confidence interval for the proportion of satisfied patients is 81.5%–88.5%.

Question 3:

A telecom company wants to estimate the average monthly data usage of its customers. A sample of 225 customers has a mean usage of 16 GB and a standard deviation of 5 GB. Construct a 99% confidence interval and interpret the result.

Solution

Given:

$$\bar{x} = 16, s = 5, n = 225$$

For a 99% confidence interval,

$$z = 2.576$$

Standard Error,

$$SE = \frac{s}{\sqrt{n}} = \frac{5}{\sqrt{225}} = 0.3333$$

Margin of Error,

$$ME = 2.576(0.3333) \approx 0.859$$

$$\therefore, CI = \bar{x} \pm ME = 16 \pm 0.859 = \boxed{(15.141, 16.859) \text{ GB}}$$

Interpretation:

We are 99% confident that the true average monthly data usage lies between 15.141 GB & 16.859 GB.

Question 4:

A company claims that the average time taken by its customer-support team to resolve a complaint is 24 hours. A sample gives a 95% confidence interval of (25.2, 27.8) hours. Based only on this confidence interval, would you reject the company's claim? Explain.

Solution

Claimed mean,

$$\mu = 24 \text{ hours}$$

Given 95% confidence interval = (25.2, 27.8)

The claimed value 24 hours does not lie within the confidence interval.

Therefore, the claim is inconsistent with the sample evidence at the 5% significance level.

Reject the company's claim

Answer

Yes. The claim of 24 hours should be rejected because 24 is outside the 95% confidence interval.

Question 5:

A manufacturer claims that the average weight of a packaged product is 500 grams. A 95% confidence interval based on a sample is (496, 502) grams. Is the claimed value of 500 grams consistent with the sample evidence? Explain your answer.

Solution

Claimed mean,

$$\mu = 500 \text{ gms}$$

Given 95% confidence interval = (496, 502)

Since $500 \in (496, 502)$

The claimed value lies within the 95% confidence interval.

<i>The claim is consistent with the sample evidence</i>

Answer

Yes. The manufacturer's claim of 500 grams is consistent with the sample evidence, since 500 lies within the 95% confidence interval.
